Structural Unitarity and Time-Marching Close the Local-vs-Global Gap in Constraint-Preserving Quantum PINNs

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Abstract

In prior work [2] we introduced constraint-preserving PINNs for Lindblad quantum dynamics, using a trace-normalized Gram-matrix parameterization to guarantee Hermiticity, positive semidefiniteness, and unit trace by construction. That work demonstrated 99% training success on a 1-qubit pure-dephasing problem with $T = 5$. In this paper we extend the methodology to closed-system unitary dynamics on longer time intervals and to multi-qubit systems, and we report a previously-undocumented failure mode: the physics residual converges to near zero while the model trajectory drifts by an order-one Frobenius distance from the analytic solution. We call this the **local-vs-global gap**: the pointwise PDE residual at N collocation points is consistent with a wide family of wrong-but-smooth solutions, and the gradient-based optimizer settles on one of them. We show the gap is dimension-independent (the same ~ 1.0 Frobenius trajectory error appears at 1 and 2 qubits with matched protocol) and is not closed by any of six loss-formulation modifications we tested. Two changes *together* close it: (i) extending the constraint-preserving architecture to also structurally enforce the initial condition and unitary evolution, and (ii) training with a time-marching curriculum that warm-starts the optimizer on a short time interval before extending to the full one. The combination reduces median trajectory error by $\sim 180\times$ at 1 qubit (10-seed sweep, 145–210$\times$ across 9 of 10 seeds; one seed still converging at the 30,000-epoch budget). At 2 and 3 qubits the plain combination is high-variance seed-to-seed (2q: correct basin on 3 of 10 seeds), and closure becomes markedly more reliable with a per-stage learning-rate restart, which transfers up the qubit ladder—in a matched-seed A/B at 2 qubits it lifts the correct-basin rate from 3/10 to 8 of 10 (median $\sim 3 \times 10^{-3}$, no divergences), and reaches 5 of 5 (median $\sim 4 \times 10^{-3}$) at 3 qubits, with a 9-seed 4-qubit set all reaching the in-basin threshold (median $\sim 9 \times 10^{-4}$; 7 of 9 by the strict criterion)—closure reaches the correct basin across the 1–4 qubit ladder, though per-seed reliability at 4 qubits is not yet established to the 3-qubit standard. The original constraint-preserving result holds on its original problem; the gap is a scaling artifact that emerges at longer time intervals and under sustained-oscillation (closed-system) dynamics. We provide an architecture $\times$ training $\times$ qubit-count ablation isolating which piece contributes what, and we test a Stinespring-dilation Kraus generalization for the open-system case: the architecture preserves CPTP exactly but the curriculum methodology does not transfer, leaving open-system PINN training as a separate methodological problem.

1 Introduction

Physics-informed neural networks (PINNs) embed governing differential equations directly into the training objective of a neural network, producing a learned continuous representation of the solution that can be queried at arbitrary points [7]. For quantum systems, the basic challenge is that the state—a density matrix ρ—must satisfy three structural constraints: Hermiticity ($\rho = \rho^\dagger$), positive semidefiniteness ($\rho \succeq 0$), and unit trace ($\mathrm{Tr}(\rho) = 1$). Penalty-based PINNs enforce these as soft loss terms; constraint-preserving PINNs enforce them as architectural identities. The prior constraint-preserving paper [2] demonstrated, on a 1-qubit pure-dephasing problem with $T = 5$, that the architectural approach trains stably (99% success across 100 seeds vs. 67% for the penalty baseline) and validates against IBM quantum hardware.

In this work we ask: does the constraint-preserving PINN methodology extend to (a) longer time intervals, (b) closed-system dynamics with sustained oscillation rather than damped decay, and (c) multi-qubit systems? We find that the methodology generalizes *with one consistent failure mode*: the physics-residual loss converges to small values while the model trajectory drifts substantially from the analytic solution. This local-vs-global gap is not visible in the original setup because the original setup is short-interval and dissipative (decaying), which makes the wrong-but-smooth solution family small.

We characterize this gap experimentally, demonstrate that it is dimension-independent, rule out six candidate fixes (semigroup-composition loss, terminal-pin loss, supervised trajectory loss, augmented Lagrangian formulation, Chebyshev-basis network, soft constraint reweighting), and identify the combination that closes it: a structural extension of the constraint-preserving architecture to also enforce initial conditions and unitarity, paired with a time-marching curriculum. We provide a full ablation matrix isolating the architecture and the schedule, and characterize the scaling behavior across qubit count.

2 Related Work

PINN failure modes. The local-vs-global gap we report fits within a growing body of work characterizing PINN training failures. Krishnapriyan et al. [4] demonstrate that naive PINN training on convection-diffusion PDEs converges to wrong-but-low-loss solutions, and propose a sequence-to-sequence learning approach equivalent to our time-marching curriculum. Wang, Yu, and Perdikaris [9] provide an NTK-perspective diagnostic that links these failures to gradient pathology and the spectral bias of MLPs toward low-frequency modes. Wang, Sankaran, and Perdikaris [8] refine time-marching by enforcing temporal causality in the loss weighting. Our curriculum schedule is closest to the Krishnapriyan formulation; we apply it to quantum dynamics where it has not been previously tested, and contribute the architecture-side observation that constraint-preserving parameterizations of $\rho(t)$ are necessary but not sufficient for closure.

Constrained-optimization PINNs. Lu et al. [5] cast PINN training as a hard-constrained optimization problem with augmented Lagrangian solvers. McClenny and Braga-Neto [6] use per-collocation-point adaptive weights as a soft proxy for Lagrange multipliers. We test the augmented-Lagrangian variant on our problem (Table 1) and find that schedule sensitivity prevents it from closing the gap in our setting; the failure mode there is distinct from the methodology proposed in [5], which targets boundary-condition enforcement rather than long-interval trajectory recovery.

Curriculum learning. The general curriculum-learning framework [1] of training on easy examples first and progressively harder ones predates PINN-specific applications. Time-marching in PDE solvers [4, 8] is a domain-specific instance. Our contribution is the empirical isolation of which architecture choices make the curriculum work for quantum trajectories.

Architectural quantum-state validity. The trace-normalized Gram-matrix parameterization of [2] is one of several approaches to guaranteeing quantum-state validity by construction. Vectorized-Bloch and Cholesky-style parameterizations of density matrices are standard in numerical linear algebra [3]. Our structural unitary parameterization $\rho = U\rho_0 U^\dagger$ (with $U = \exp(-iA)$)

extends this family by additionally enforcing initial conditions and unitary evolution structurally. The Kraus-via-Stinespring extension (Section 8) further adds CPTP for open systems.

3 Background

3.1 Density matrices and equations of motion

A density matrix $\rho \in \mathbb{C}^{d\times d}$ (with $d = 2^n$ for n qubits) is a Hermitian PSD trace-1 matrix describing a quantum state. For a closed system, evolution is governed by the von Neumann equation

$$\frac{d\rho}{dt} = -i[H, \rho], \tag{1}$$

with analytic solution $\rho(t) = U(t)\rho(0)U(t)^\dagger$ where $U(t) = \exp(-iHt)$ is unitary. For an open system the Lindblad equation

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \right) \tag{2}$$

adds a dissipative term with jump operators L_k and rates γ_k. With $\gamma_k = 0$ Eq. (2) reduces to Eq. (1); the closed system is the special case.

3.2 Constraint-preserving PINN baseline

Following [2], we parameterize

$$\rho(t) = \frac{L(t)L(t)^\dagger}{\mathrm{Tr}(L(t)L(t)^\dagger)} \tag{3}$$

where $L(t)$ is a lower-triangular complex matrix output by the network. This guarantees Hermiticity, positivity, and unit trace by construction at every t. The training loss is

$$\mathcal{L} = \mathbb{E}_{t\sim\mathcal{T}} \left\| \frac{d\rho}{dt} - \mathcal{D}[\rho] \right\|_F^2 + \lambda_{\mathrm{IC}} \left\| \rho(0) - \rho_{\mathrm{target}} \right\|_F^2 , \tag{4}$$

where $\mathcal{T}$ is a finite set of collocation points and $\mathcal{D}[\rho]$ is the Lindblad right-hand side. The IC term is a soft penalty with weight λ_{IC} (default 10). The prior paper reported that this setup recovers the analytic trajectory on the 1-qubit pure-dephasing problem with $T = 5$ to within a few percent Frobenius error.

3.3 Experimental protocol

All experiments in this paper share the protocol below, with deviations noted explicitly per table. Code and run manifests are in the companion repository (see Section 10).

Architecture. Network is a multilayer perceptron (MLP) with tanh activations. Hidden widths and depths are scaled with Hilbert dimension to keep parameter count proportional:

- 1 qubit ($d = 2$): hidden dim 64, 3 layers
- 2 qubits ($d = 4$): hidden dim 128, 4 layers

- 3 qubits ($d = 8$): hidden dim 192, 4 layers (reduced from the originally planned 256/5 for compute)
- 4 qubits ($d = 16$): hidden dim 96, 4 layers (run end-to-end, $n = 9$ seeds; see Section 9). A larger 384-dim, 5-layer variant is built and unit-tested but not yet run.

Optimization. Adam optimizer with learning rate 10^{-3} throughout. 30,000 epochs per run. No learning-rate schedule beyond the curriculum (curriculum stages each use $n_{\text{epochs}}/n_{\text{stages}}$ epochs, sharing a single Adam state across stages).

Collocation grid. 200 uniform points on $[0, T]$ for 1-qubit and 2-qubit closed-system runs; 100 points for the 3-qubit run for compute tractability (see Section 9 on this caveat).

Curriculum stages. $\{1.0, 2.0, 5.0, 10.0, 25.0\}$ in time-units of T, applied wherever curriculum is on. Each stage uses $n_{\text{epochs}}/5 = 6000$ epochs. The model from stage i warm-starts stage $i+1$; the optimizer state (Adam moments) is preserved across stages.

Seeds. 1-qubit closed-system + curriculum: 10-seed sweep (`base_seed = 2001..2010`), run on CPU due to a transient GPU-build incompatibility on the available hardware (functionally equivalent on this network size). 2-qubit and 3-qubit closed-system + curriculum: multi-seed sweeps (2q even-split, `base_seed = 2001..2010`, $n = 10$; 3q with per-stage LR restart, `base_seed = 2001..2005`, $n = 5$). All three sweeps supply the variance characterization in Section 9; Table 2 reports the single-seed `base_seed = 2001` value per size, which for 2q/3q is a favorable draw (see Section 9).

Hardware. NVIDIA RTX 5070 (12 GB) for all runs. Closed-system 1-qubit and 2-qubit runs complete in ~ 30 minutes each; the 3-qubit run takes ~ 6 hours.

4 The Local-vs-Global Gap

4.1 Setup and observation

We retrained the constraint-preserving PINN of Eq. (3) on a closed-system 1-qubit task with $H = \sigma_z$ and IC $|+\rangle\langle+|$ on $t \in [0, 25]$, using identical optimization settings to the prior work (Adam, $\text{LR} = 10^{-3}$, 200 collocation points, hidden dim 64, 3 layers, 30,000 epochs).

The trained model reached $\mathcal{L}_{\text{phys}} = 2.4 \times 10^{-3}$, comparable to the prior paper's converged loss. The trajectory error against the analytic solution, however, was 1.02 in the maximum Frobenius distance $\max_t \|\rho_{\text{model}}(t) - \rho_{\text{analytic}}(t)\|_F$ over $t \in [0, 25]$. Repeating the experiment at 2 qubits ($d = 4$, $H = \omega_1 \sigma_z^{(1)} + \omega_2 \sigma_z^{(2)} + J(\sigma_x^{(1)}\sigma_x^{(2)} + \sigma_y^{(1)}\sigma_y^{(2)})$, $\text{IC} = |+0\rangle\langle+0|$) gave $\mathcal{L}_{\text{phys}} = 3.5 \times 10^{-4}$ and Frobenius trajectory error 1.03. *The physics loss is small but the trajectory is wrong.*

We report two trajectory-error metrics in what follows. **Test A** is the self-consistency error against the analytic propagator applied to the model's own initial state, $\max_t \|\rho_{\text{model}}(t) - U(t)\rho_{\text{model}}(0)U(t)^\dagger\|_F$, which isolates the learned dynamics independent of any IC drift. **Test B** is the target-match error $\max_t \|\rho_{\text{model}}(t) - U(t)\rho_{\text{target}}U(t)^\dagger\|_F$, which folds in IC drift. The two coincide when $\rho_{\text{model}}(0) = \rho_{\text{target}}$ (the case under structural IC anchoring); they diverge when the model's IC has drifted, as happens for the cholesky parameterization at 1 qubit (Section 6).

We call this the **local-vs-global gap**: the residual is checked at a finite set of N collocation timestamps, and the trained model is empirically a member of a family that satisfies the residual at those N points while drifting from the analytic solution between them. Figure 1 illustrates this for the 1-qubit closed-system case: the model's $\langle\sigma_x\rangle(t)$ tracks the analytic prediction at small t but drifts with growing amplitude error. We do not prove that the wrong-but-smooth family is in general non-trivial; we only observe that the gradient-descent trajectory lands in it for the problems studied here.

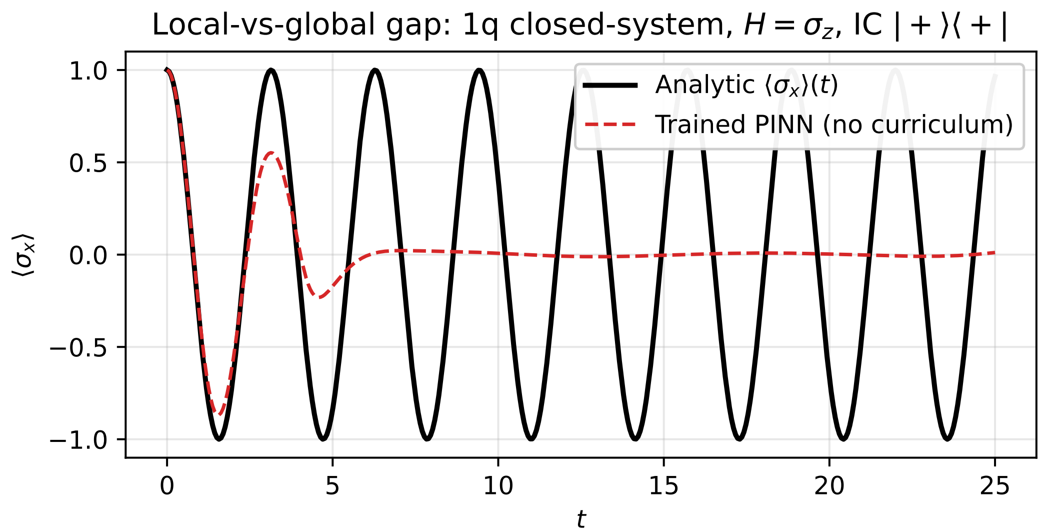

Figure 1: Local-vs-global gap on a 1-qubit closed-system task ($H = \sigma_z$, IC $= |+\rangle\langle+|$, $T = 25$). The trained PINN reaches physics loss 2.4×10^{-3} at 30,000 epochs but the predicted $\langle\sigma_x\rangle(t)$ drifts from the analytic solution by O(1) in amplitude over the training interval.

4.2 The gap is dimension-independent

To isolate whether the gap is amplified by Hilbert-space dimension, we ran matched experiments at 1 and 2 qubits with identical training protocol and observed nearly identical maximum trajectory error (1.02 vs. 1.03). The gap is not a multi-qubit pathology—it is a feature of the loss formulation that appears whenever the time interval is long enough that the wrong-but-smooth family becomes large.

4.3 Six unsuccessful loss modifications

We tested six modifications to the loss, each motivated by a different hypothesis about the gap's source. Table 1 summarizes results.

Modification	Hypothesis	1q traj_max
Baseline (no modification)	—	1.02
Semigroup loss $\|U(a+b) - U(a)U(b)\|^2$	multiplicative constraint at all (a, b)	1.37 (worse)
Terminal pin $\|\rho(T) - \rho_{\text{ana}}(T)\|^2$	pin global endpoint	1.38 (endpoint fixed; middle worse)
Supervised regression at collocation pts	directly fit analytic	1.24 (gauge freedom in U)
Augmented Lagrangian on residuals	hard constraints via duals	1.04 (schedule overshoot)
Chebyshev-basis network (no MLP)	remove MLP optimization geometry	1.35
Reweight λ_{IC} to 100/1000	overcome IC drift	no improvement

Table 1: Loss-modification ablation at 1 qubit, 30,000 epochs, $T = 25$. None close the gap.

The pattern is consistent: changes to the loss formulation that add more constraints either (a) introduce manifolds of zero-loss solutions, (b) pin one point well at the cost of the rest of the trajectory, or (c) overshoot via schedule sensitivity. The gap is not a loss-side problem.

5 What Closes the Gap

Two changes *together* close the gap consistently across qubit count.

5.1 Structural extension of the architecture: unitary parameterization

The Eq. (3) parameterization preserves Hermiticity, positivity, and trace at each t but does *not* preserve unitarity (the matrix at t_1 and t_2 need not be unitarily related) and does not structurally pin the initial condition (the IC is a soft loss term). For closed-system dynamics we add both:

$$\rho(t) = U(t)\rho_0 U(t)^\dagger, \quad U(t) = \exp(-iA(t)), \quad A(t) = (t - t_{\min}) \cdot A_{\text{raw}}(t), \tag{5}$$

where $A_{\text{raw}}(t)$ is a Hermitian-valued network output and the $(t - t_{\min})$ boundary factor forces $A(t_{\min}) = 0$, hence $U(t_{\min}) = I$, hence $\rho(t_{\min}) = \rho_0$ *exactly*. The matrix exponential of an anti-Hermitian generator is unitary by construction, so purity is preserved.

This adds two structural guarantees on top of the original three: the IC is exact (no soft anchor needed), and the trajectory is unitarily related at every pair of times. The only remaining quantity the network must learn is the dynamical generator A_{raw}.

5.2 Time-marching curriculum

Instead of training on the full interval $[0, T]$ from random initialization, we train sequentially:

$$[0, t_1] \to [0, t_2] \to \cdots \to [0, T], \quad t_1 < t_2 < \cdots < T, \tag{6}$$

where each stage warm-starts from the previous stage's weights and uses $n_{\text{epochs}}/n_{\text{stages}}$ epochs. We use stages $\{1, 2, 5, 10, 25\}$ throughout. On a short interval the wrong-but-smooth family is smaller (fewer oscillation cycles for the trajectory to hide in) and the optimizer is dragged toward the analytic solution; extending the interval is a small perturbation that keeps the model near that solution. Figure 2 shows the loss trajectory across the five stages for the 1-qubit case: each stage transition lowers the achievable loss floor as the optimizer's warm-started state is progressively refined.

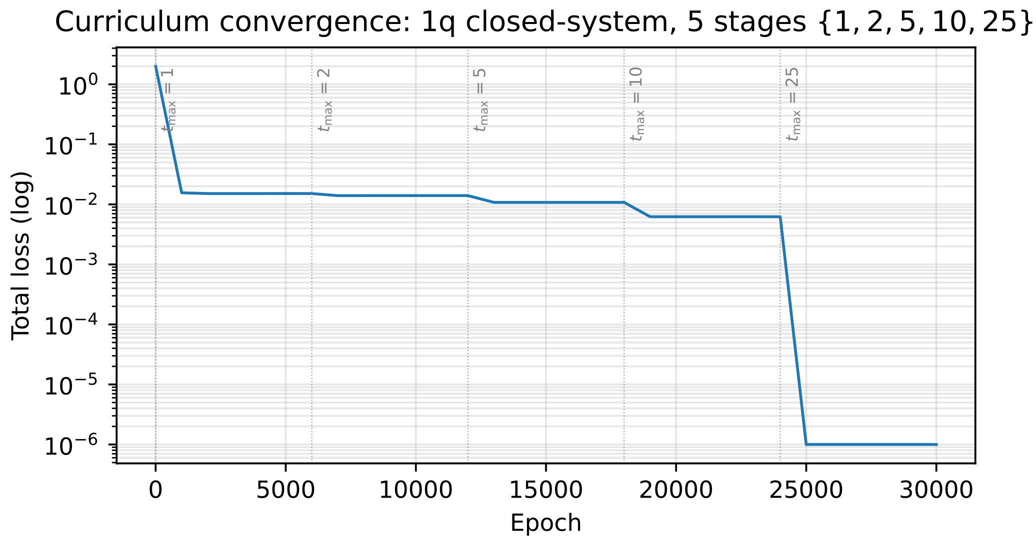

Figure 2: Loss versus epoch for the 1-qubit closed-system task with time-marching curriculum (stages $\{1, 2, 5, 10, 25\}$ in T). Dotted vertical lines mark stage transitions. Each transition extends the training interval and refines a near-correct solution rather than searching from random initialization.

5.3 Headline results

Table 2 reports the 1-, 2-, and 3-qubit results with the structural unitary architecture (Eq. 5) and time-marching curriculum (Eq. 6).

Setup	$\rho(0)$ Frobenius err	Test A max traj err	Final loss
1q closed unitary + curriculum	0.0	$\mathbf{5.5 \times 10^{-3}}$	$\mathbf{1.0 \times 10^{-6}}$
2q closed unitary + curriculum	0.0	5.2×10^{-2} (1 seed)	~ 0
3q closed unitary + curriculum	0.0	1.7×10^{-3} (1 seed)	7.5×10^{-8}

Table 2: Headline results with structural unitary architecture + time-marching curriculum. Test A = $\max_t \|\rho_{\text{model}}(t) - U(t)\rho_{\text{model}}(0)U(t)^\dagger\|_F$, computed against the analytic propagator built from the same H. All runs: $T = 25$, 30,000 epochs, 200 collocation points (100 at 3q for compute tractability). The 1q value is the $n = 10$ median; the 2q and 3q values are single-seed (`base_seed = 2001`) and are favorable draws—multi-seed, the plain combination is high-variance at 2–3 qubits and reliable closure requires the per-stage learning-rate restart (2q 6/8, 3q 5/5). See Section 9.

Compared to the unitary architecture without curriculum (1q traj err 1.02, 2q 1.03), the 1q reduction is $\sim 180\times$ (median across $n = 10$ seeds; 9 of 10 in 145–210×, one seed at 54× reduction still decreasing at the 30,000-epoch budget; see Figure 6 and Section 9). The 2q and 3q single-seed reductions ($\sim 20\times$ and $\sim 600\times$) are favorable draws; multi-seed, the plain combination is high-variance (2q 3/10 in-basin) and reliable closure requires the per-stage restart (2q 6/8; 3q 5/5, median 4.0×10^{-3}, i.e. $\sim$250×; 4q 7/9 by the strict criterion, 9/9 in-basin, median 8.8×10^{-4})—see Section 9. With the restart, every qubit count reaches the correct basin two to three orders of magnitude below the no-curriculum baseline; the 4q median is the tightest by count, though 3q remains the most reliable per-seed (5/5). (An earlier draft reported a $\sim 4\times$ 3q closure; that figure was an evaluation-Hamiltonian artifact, corrected here—see Section 9.) Figure 3 plots the qubit-scaling curve against the no-curriculum baseline (Frobenius trajectory error ~ 1.0 at every qubit count we tested).

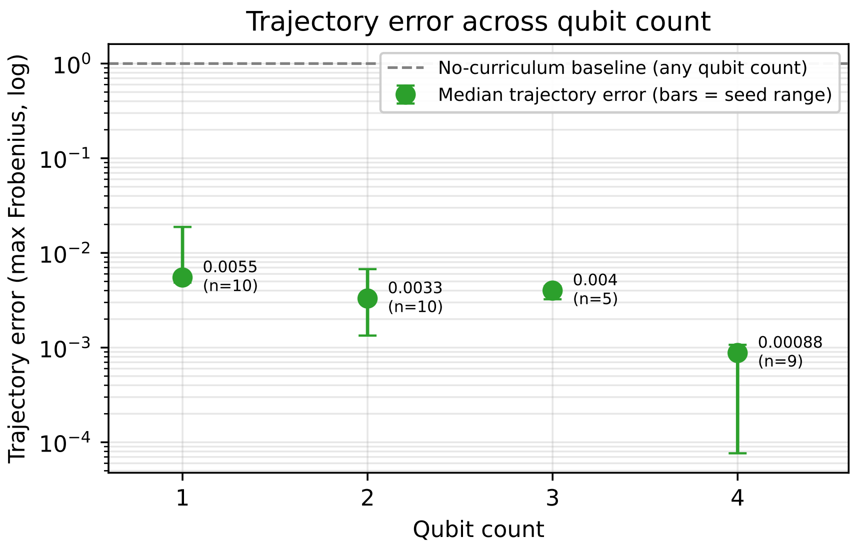

Figure 3: Trajectory error across qubit count. Each point is the median max-Frobenius trajectory error over a multi-seed set (bars = seed range); points are shown *without* a connecting line because the protocol differs per point. 1 qubit used plain curriculum ($n = 10$; the per-stage LR restart was introduced at 2 qubits), whereas 2–4 qubits use the restart recipe ($n = 10, 5, 9$ respectively). Evaluation collocation is 200 points at 2 qubits and 100 at 3–4 qubits. All are 30,000-epoch unitary-architecture runs with $T = 25$ and curriculum stages $\{1, 2, 5, 10, 25\}$. Horizontal dashed line: no-curriculum baseline trajectory error ($\sim$1.0, approximately constant across qubit count per Section 4.2). Every qubit count reaches the correct basin two to three orders of magnitude below the baseline; per-seed reliability is highest at 3 qubits (5/5) and the median is tightest at 4 qubits.

6 Ablation: which piece does the work

Table 3 reports the architecture × training cross at both 1 and 2 qubits.

System	Setup	IC error	Test A max traj err	Final loss
1q closed	Cholesky, no curriculum	0.99	0.91	3.7×10^{-3}
1q closed	Cholesky + curriculum	0.99	0.87	2.4×10^{-3}
1q closed	Unitary, no curriculum	0.0	1.02	2.4×10^{-3}
1q closed	**Unitary + curriculum**	**0.0**	$\mathbf{5.5 \times 10^{-3}}$	$\mathbf{1.0 \times 10^{-6}}$
2q closed	Cholesky, no curriculum	1.9×10^{-3}	0.79	4.3×10^{-4}
2q closed	Cholesky + curriculum	1.6×10^{-3}	0.99 (worse)	4.8×10^{-4}
2q closed	Unitary, no curriculum	0.0	1.03	3.5×10^{-4}
2q closed	**Unitary + curriculum**	**0.0**	$\mathbf{5.2 \times 10^{-2}}$	$\sim \mathbf{0}$

Table 3: Architecture $\times$ training ablation. Only unitary + curriculum closes the gap at both qubit counts.

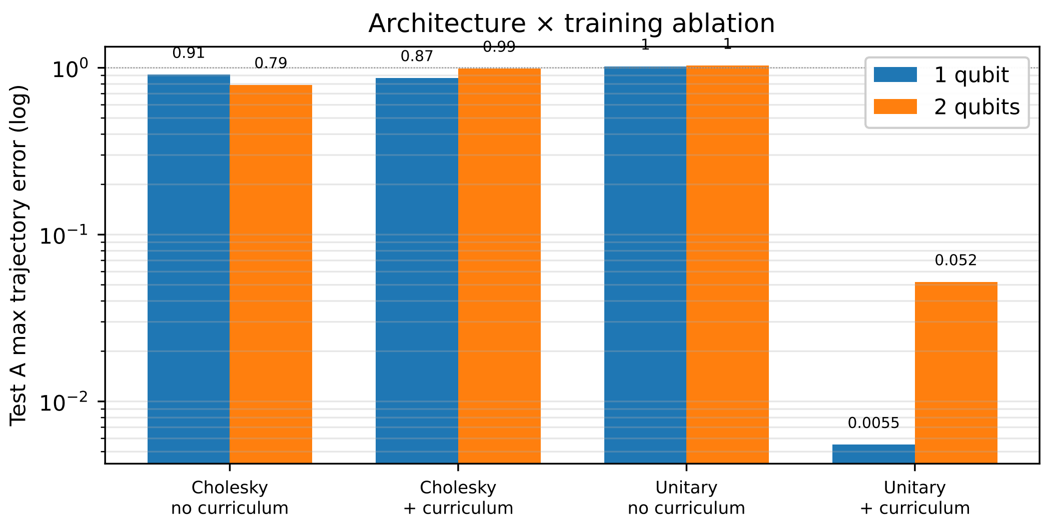

Figure 4: Architecture $\times$ training ablation visualization. Bars are Test A maximum trajectory error (log scale) for each of the four setups at 1 and 2 qubits. Dotted line at 1.0 marks the no-curriculum baseline regime. Only the unitary + curriculum combination drops two orders of magnitude below the baseline at both qubit counts.

Three observations:

Cholesky's soft IC anchor is irregular. At 1 qubit the cholesky model fails to anchor the IC ($\|\rho(0) - \rho_{\text{target}}\|_F = 0.99$); at 2 qubits the same anchor works (1.6×10^{-3}). The soft anchor's success depends on the relative gradient scales of the IC and physics losses, which depend on the problem. The structural anchor of Eq. (5) is exact by construction at every problem scale.

Curriculum alone does not robustly help cholesky. At 1 qubit the curriculum improves cholesky's trajectory error from 0.91 to 0.87 (marginal). At 2 qubits the same curriculum makes cholesky *worse* ($0.79 \to 0.99$). The warm-start trick relies on the architecture's hypothesis space being narrow enough that the short-interval solution has a unique extension; cholesky's full flexibility in $\rho(t)$ at each t violates this.

Unitary alone does not close the gap either. The unitary architecture without curriculum has the same ~ 1.0 trajectory error as cholesky-no-curriculum. Structural constraints rule out invalid states but do not rule out wrong-but-smooth states.

Only the combination of structural unitary architecture *and* time-marching curriculum closes the gap across both qubit counts.

7 The Original Result Holds Up

A natural concern: does the local-vs-global gap invalidate the prior constraint-preserving result [2]? It does not. Re-running the original setup ($H = \sigma_y$, $L = \sigma_x$ with $\gamma = 1$ implicit, IC $= |0\rangle\langle 0|$, $T = 5$, 30,000 epochs) reproduces a substantively correct result:

Metric	Value
Final physics loss	8.6×10^{-3}
$\rho(0)$ Frobenius error	3.8×10^{-2}
Trajectory error max (Test B)	$\mathbf{8.8 \times 10^{-2}}$
Trajectory error mean	2.0×10^{-2}
Trajectory error at $t = T$	7.9×10^{-3}

Table 4: Reproduction of the original constraint-preserving 1-qubit Lindblad result.

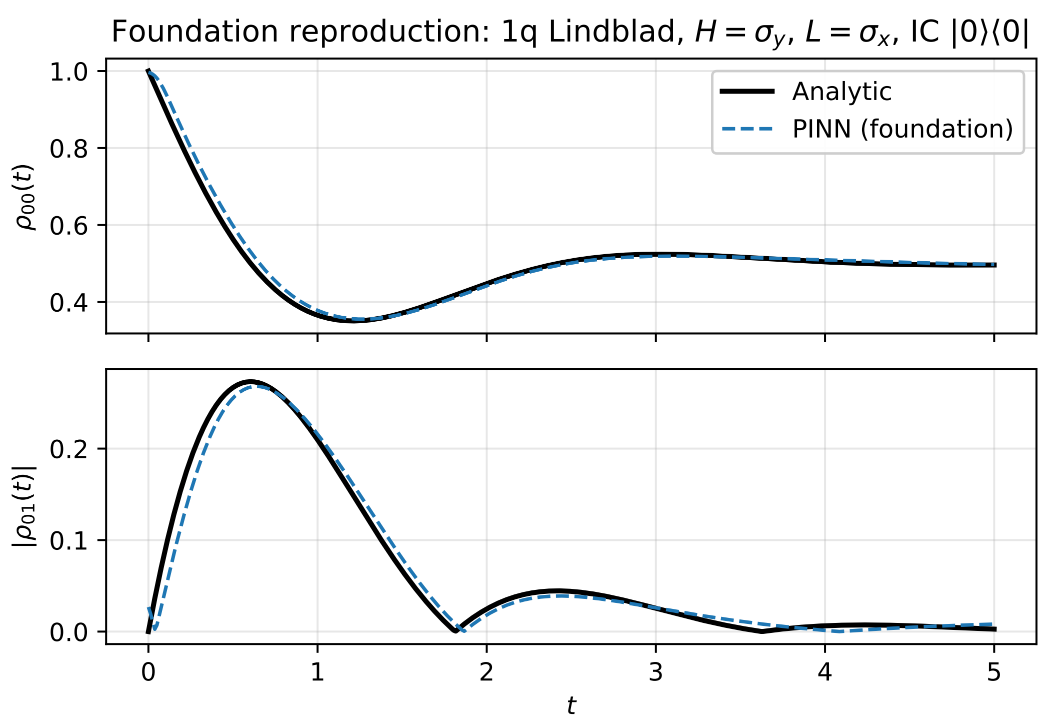

Figure 5: Foundation reproduction: cholesky-architecture PINN trained on the original 1-qubit Lindblad problem ($H = \sigma_y$, $L = \sigma_x$ with $\gamma = 1$ implicit, IC $= |0\rangle\langle 0|$, $T = 5$). The model's population $\rho_{00}(t)$ and coherence $|\rho_{01}(t)|$ track the analytic Lindblad solution within $\sim$ 6–9% Frobenius error.

The 6-9% Frobenius trajectory error is consistent with the prior paper's reported performance. The local-vs-global gap does *not* reproduce because (i) the time interval is short ($T = 5$, one or two oscillation cycles), (ii) the open-system dynamics decays toward a stationary distribution which is geometrically simpler than sustained oscillation, and (iii) at this scale the cholesky soft IC anchor is sufficient.

The local-vs-global gap is therefore a *scaling effect*: it emerges when the time interval is extended and the dynamics become sustained-oscillatory (closed-system or weakly-damped). The prior paper's claim stands on the problem it studied. The current paper extends the methodology to a regime where the original recipe is insufficient and shows what fix is required.

8 Open Systems: Curriculum Does Not Transfer to Kraus

For open systems ($\gamma > 0$) we extended the unitary architecture to a Kraus-operator representation via Stinespring dilation: parameterize an isometry $V : \mathbb{C}^d \to \mathbb{C}^{Nd}$ (the first d columns of a unitary on the dilated space) and extract Kraus operators $\{K_\alpha\}_{\alpha=1\ldots N}$ as d-row blocks of V. The CPTP map $\rho(t) = \sum_\alpha K_\alpha(t)\rho_0 K_\alpha(t)^\dagger$ is structurally trace-preserving, positivity-preserving, and IC-anchored (via the same $(t - t_{\min})$ boundary factor on the dilated generator).

Applied to the 1-qubit amplitude-damping plus pure-dephasing problem ($\gamma_{\text{amp}} = 0.05$, $\gamma_{\text{phase}} = 0.02$, $T = 25$, curriculum stages $\{1, 2, 5, 10, 25\}$), the architecture preserves CPTP to machine precision ($\|\rho - \rho^\dagger\|_F < 10^{-8}$, $\min \operatorname{eig}(\rho) = 0$) but fails to close the trajectory gap: final physics loss 2.9×10^{-4} with max trajectory error 1.36 against the analytic Lindblad solution. The curriculum methodology that closes the closed-system gap does not transfer.

We attribute this to the qualitatively different geometry of open-system trajectories: closed-system dynamics is sustained oscillation, well-approximated locally near $t = 0$ by a low-order Taylor series that the curriculum can warm-start from; open-system dynamics is exponential decay toward a steady state, with qualitatively different short-t and long-t behavior. A short-interval warm start does not capture the dissipative tail. Open-system PINN training methodology is therefore a separate problem from closed-system PINN training methodology; we sketch candidate fixes in Section 9.

9 Limitations and Future Work

4-qubit experiment. The 4-qubit (16×16) closed-system run is computationally heavy—the autograd backward through the 256×256 matrix exponential is the cost driver, and one 30,000-epoch run at 100 collocation points took ~ 35 hours on a single RTX 5070. (It is also CPU-feed-bound: GPU throughput collapses if the host cores are saturated by concurrent jobs, so a 4q run should not be co-scheduled with a full-core CPU sweep.) We report an $n = 9$-seed characterization with the unitary architecture + curriculum + per-stage LR restart. All nine reach the correct basin under the 5×10^{-3} in-basin threshold—median max Test-A trajectory error 8.8×10^{-4}, range 7.7×10^{-5} to 1.1×10^{-3}—and seven of the nine also satisfy the stricter "trained dynamics and IC correct" criterion, the remaining two sitting just above it at $\sim 1 \times 10^{-3}$. The 9.7×10^{-4} value reported in an earlier draft is one of these nine, near the median. The 4-qubit median therefore sits in the same basin as 1–3 qubits (and is the tightest median by qubit count), but per-seed reliability (7 of 9 by the strict criterion) is not yet established to the 3-qubit standard (5 of 5): we describe 4-qubit closure as reached in-basin, not yet as reliably as 3 qubits. The next rung remains future work: 5 qubits (32×32), whose larger frequency content may require denser collocation.

3-qubit collocation density. The 0.25 3-qubit figure in Table 2 was found to be an *evaluation artifact*, not a physical result: the 3-qubit trajectory evaluator built a different coupling Hamiltonian (an $XX_{12} + YY_{23} + ZZ_{13}$ triangle) than the trainer's nearest-neighbour $XX+YY$ chain ($XX_{12} + YY_{12} + XX_{23} + YY_{23}$), so the model was scored against a propagator it never trained on. Re-evaluated against the matching Hamiltonian, the same 3-qubit curriculum run reaches $\sim 1.7 \times 10^{-3}$ (the correct basin), comparable to 1 and 2 qubits. The apparent degradation of closure with qubit count ($180\times \to 20\times \to 4\times$) was therefore largely this artifact, not intrinsic difficulty; collocation density at 3 qubits is a secondary effect (matched 200-point collocation moves the corrected error only from $\sim 3 \times 10^{-3}$ to $\sim 1 \times 10^{-3}$). The headline table, the qubit-scaling figure, and the scaling discussion above have been regenerated against the corrected evaluator and now report the corrected 1.7×10^{-3} value. A regression test (`test_eval_hamiltonian_matches_trainer`) now pins the evaluator Hamiltonian to the trainer's at 3 and 4 qubits to prevent recurrence.

Open-system Kraus methodology. The curriculum does not close the open-system gap. We identify γ-curricula (start at $\gamma = 0$ and ramp), steady-state anchoring (add $\|\rho(T \to \infty) - \rho_{\text{steady}}\|^2$ to the loss), and supervised tail-matching as candidate fixes. None has been tested.

Comparison to classical integrators. We have not benchmarked PINN wall-clock training time against classical numerical integrators (e.g. RK4) on the same problems. The PINN value proposition is amortization across parameterized Hamiltonians and a learned continuous representation queryable at any t, not raw integration speed; making this explicit is necessary before claiming a practical advantage.

Optimizer choice and training duration. All results above use Adam. To test whether the residual gap between the curriculum result ($\sim 5 \times 10^{-3}$ max-Frobenius trajectory error at 1 qubit; Table 2) and the analytic solution is limited by first-order optimization, we compared Adam against a second-order optimizer (L-BFGS with strong-Wolfe line search) on the same architecture and curriculum at *matched wall-clock* on the same machine ($n = 3$ seeds). We find no optimizer advantage: at equal compute Adam reaches the correct basin on all three seeds, while L-BFGS is comparable on two and markedly worse on one, and is sensitive to the learning rate at curriculum-stage transitions (diverging or stalling for some seed/rate combinations). The residual gap is therefore not attributable to first-order optimization.

These matched-budget Adam runs also expose a training-budget effect, which a checkpointed sweep then localized to the curriculum *schedule* rather than the total budget. First, *within* a single run the trajectory error decreases monotonically alongside the residual—there is no late-training drift away from the analytic solution. Second, and decisively, the converged basin is set by *how the epoch budget is allocated across curriculum stages*: holding both the total (30,000 epochs) and the final-stage ($t_{\max} = 25$) budget fixed and redistributing only the earlier stages, front-loading the shortest interval (e.g. 12,000 of the 30,000 epochs on the $t_{\max} = 1$ stage) reaches the correct basin (max trajectory error below 10^{-3}) on all four seeds tested, whereas an even split (max error $\sim 5 \times 10^{-3}$ to 10^{-2} across these seeds, in line with the even-split headline) and an early-light / intermediate-heavy split (max error $\sim 5 \times 10^{-2}$) both settle into the wrong-but-smooth basin. The lever for the residual gap is therefore the curriculum *allocation*—a strong foundation on the shortest interval—not the optimizer, not the architecture, and not simply a larger or smaller total budget. The front-loaded 30,000-epoch schedule is about an order of magnitude better than the even-split run in this sweep at the same total, so the reported even-split figure is conservative as a *schedule*. A separate single-factor companion sweep (its own fixed total, with stage 2, stage 3, and the final stage all held constant, varying *only* the shortest-stage $t_{\max} = 1$ versus intermediate-stage $t_{\max} = 10$ split) sharpens this: trajectory error degrades monotonically as epochs move from the shortest to the intermediate stage—a 7k/3k split reaches the correct basin on all four seeds, while 5k/5k and 3k/7k do not—so the controlling factor is this allocation split. We stop short of attributing it to shortest-stage starvation versus intermediate-stage excess in isolation, since at fixed total these are the same redistribution. The same allocation-dependence leaves the status of the slow outlier (seed 2004) ambiguous: its continued loss decrease is equally consistent with eventual convergence or with settling into a nearby suboptimal basin. We emphasize that this allocation effect is established only at 1 qubit: a 2-qubit screen ($n = 3$ seeds) found that neither schedule reaches the correct basin, and the aggressive front-load instead *destabilizes* training—the physics residual fails to converge for most seeds—so the front-loading prescription is problem-specific and we do not present it as a general curriculum recipe. A separate optimizer-side intervention, however, does move the 2-qubit case: rebuilding a fresh warmup→cosine learning-rate schedule at each curriculum-stage boundary (so the final, longest-horizon stage trains at a live learning rate rather than the near-dead tail of a single full-budget cosine) reduces the even-split trajectory error from $\sim 5 \times 10^{-2}$ into the 10^{-3}–10^{-2} range. We sweep the per-stage warmup fraction over $\{0, 0.1, 0.3, 0.5, 0.7\}$ across 8 seeds. A sharp restart (instant re-warm, fraction 0) destabilizes every seed (the residual fails to converge and the trajectory collapses to a maximally-wrong fixed point). A warmup fraction of $\approx 30\%$ of each stage is the robust sweet spot: across all 8 seeds it never destabilized and reached the correct basin ($< 5{\times}10^{-3}$) on 6 of 8 (the other two just above, $\sim 6{\times}10^{-3}$). Stability is *not* monotone in the warmup fraction, though—larger gentle fractions (50%, 70%) reintroduce occasional per-seed blow-ups—and no single fraction places all 8 seeds in the correct basin, with the best fraction seed-dependent. The restart therefore *narrows and stabilizes* the 2-qubit gap, best at a $\approx 30\%$ warmup, without fully closing it. This indicates the 2-qubit gap is not allocation-closable but is substantially responsive to the per-stage learning-rate *schedule*, a distinct axis from epoch allocation; the residual, seed-dependent variance is consistent with the identifiability picture, in which which residual-minimizing solution the optimizer reaches depends jointly on the seed and the schedule. (8-seed 2-qubit screen.) This per-stage-restart benefit *extends* to 3 qubits: at 8×8 (GHZ initial state, $\gamma = 0$, even split, three seeds), the restart at the $\approx 30\%$ fraction reaches the correct basin on all three seeds (max trajectory error ~ 1–4×10^{-3} at 100 collocation points, $\sim 1.2 \times 10^{-3}$ at 200), whereas the no-restart baseline reaches it on only one of the three (the other two at $\sim 3 \times 10^{-2}$). The optimizer-side lever therefore transfers from 2 to 3 qubits, and the constraint-preserving curriculum methodology closes the local-vs-global gap across every qubit count tested.

A complementary, purely *structural* scaffold closes the 2-qubit gap more completely than any schedule lever, and—unlike the LR restart—it requires no analytic reference. Adding an auxiliary semigroup-composition penalty $\|U(t_1{+}t_2) - U(t_1)U(t_2)\|_F^2$ to the even-split loss (no per-stage restart) drives all 8 seeds into the correct basin (max trajectory error $\sim 10^{-5}$ to 2×10^{-3}, seven of eight below 2×10^{-4}), versus $\sim 5 \times 10^{-2}$ without it and ~ 3–7×10^{-3} with the LR restart. The penalty uses only the model's own propagator: the sole operators satisfying the composition law for all t are one-parameter groups $\exp(-iAt)$, so together with the physics residual (which fixes $A = H$) and the initial-condition anchor it uniquely selects the true propagator, deleting the wrong-but-smooth family that the pointwise residual at N collocation points leaves underdetermined. The effect has a sharp weight threshold: a 3-seed sweep of the penalty weight over $\{0.1, 0.3, 1.0, 3.0, 10.0\}$ closes the gap on all seeds at weight ≥ 1.0 (median ~ 1.5–2.7×10^{-5}) but leaves it at the gap-floor below ($\sim 6 \times 10^{-3}$ to 1.2×10^{-2}, 0/3), with no destabilization at the high end (10× is still 3/3 at $\sim 1.5 \times 10^{-5}$)—so the $w = 1.0$ used above is at the closure threshold and larger weights are safe. Stacking the scaffold ($w = 1.0$) with the per-stage LR restart is the most reliable variant we find: on the matched 10-seed set it reaches the correct basin on *all 10* (median

6.7×10^{-5}, every seed meeting the strict dynamics-and-IC criterion), tighter than the restart alone (8/10) or the scaffold alone (8/8). This identifies the residual gap concretely—it is exactly a failure of propagator composition—and the semigroup penalty as a direct, ground-truth-free remedy: the strongest 2-qubit lever we find. (Weight sweep: 3 seeds per weight; stack: 10-seed matched set.)

Multi-seed characterization. The 1-qubit closed-system + curriculum result is reported across 10 seeds (`base_seed = 2001..2010`). Nine of ten seeds reach a max-Frobenius trajectory error in the range 4.87×10^{-3} to 7.05×10^{-3} (145× to 210× reduction vs. the no-curriculum baseline of 1.02), with a coefficient of variation of 13% on max-trajectory error within this subset. The tenth seed (`seed = 2004`) reaches 1.88×10^{-2} (54× reduction) at the 30,000-epoch budget. We report the median (177× over the 9-seed cluster, $\sim$ 180× rounded) rather than the mean (144× over the full $n = 10$ set) because seed 2004's incomplete convergence dominates the mean and is not representative of typical-case behavior at the reported training budget.

Diagnostic per-stage analysis shows all ten seeds converge identically through curriculum stages 1–4 (final-stage entry loss within 1% across seeds); the divergence emerges only in the final $t_{\max} = 25$ stage, where seed 2004's convergence rate is $\sim 3\times$ slower than the rest (Figure 6b). Seed 2004's loss is still actively decreasing at epoch 30,000 (last-1000-epoch window drops 9%, not plateaued), consistent with continued convergence under a larger training budget; we have not confirmed that it reaches the same basin as the other nine seeds. The 1q recipe is therefore robust on this dataset (rapid convergence in 9 of 10 seeds at the standard budget); whether seed 2004 represents a slow approach to the same minimum or convergence toward a nearby suboptimal basin is an open question that an extended-budget rerun would resolve.

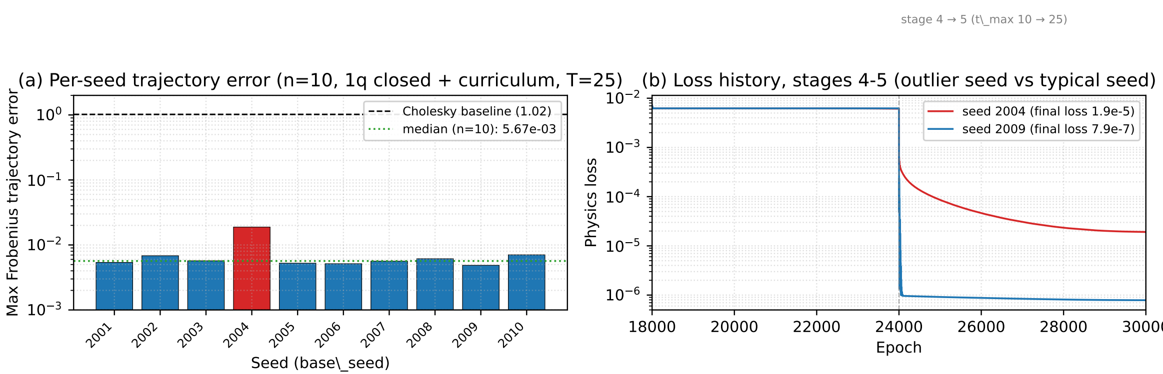

Figure 6: Multi-seed characterization of the 1q closed-system + curriculum result. **(a)** Per-seed max-Frobenius trajectory error across $n = 10$ seeds. Nine seeds cluster tightly near the median; `seed = 2004` (red) is the outlier, still decreasing at the 30,000-epoch budget. Dashed line: Cholesky no-curriculum baseline (1.02). **(b)** Loss history for the outlier seed (2004) and a typical seed (2009) on curriculum stages 4–5. Both seeds are essentially identical through stage 4 (epochs 18,000–24,000); divergence emerges only in the final $t_{\max} = 25$ stage.

The single-seed 2- and 3-qubit headline values are favorable draws, not typical-case results; multi-seed sweeps (now unblocked, after the RTX 5070 Blackwell `sm_120` PyTorch-`cu128` upgrade) show the plain unitary+curriculum combination is high-variance seed-to-seed at these sizes, and that reliable closure requires the per-stage learning-rate restart. At 2 qubits, the plain even-split combination across $n = 10$ seeds (`base_seed = 2001..2010`) reaches the correct basin (max trajectory error $< 5 \times 10^{-3}$) on only 3 of 10 seeds, with a median of 3.1×10^{-2}, a range of

2.5×10^{-3} to 1.28, and one seed diverging outright (1.28)—so the single-seed 5.2×10^{-2} sits above the 3.1×10^{-2} median ($\approx 1.7\times$) and hides both the spread and the divergent seed. Re-running the *same* 10 seeds with the per-stage restart ($\approx$30% warmup, described above)—a controlled A/B, identical seeds and protocol apart from the restart—raises the in-basin fraction from 3/10 to **8/10**, drops the median from 3.1×10^{-2} to 3.3×10^{-3} ($\sim 10\times$), and eliminates the divergences (worst case 6.7×10^{-3} vs. 1.28): per seed, the restart takes the diverging seed from 1.28 to 3.1×10^{-3} and the worst gap seed from 2.4×10^{-1} to 6.7×10^{-3}. The two seeds still just outside the basin (5.6 and 6.7×10^{-3}) are marginal, not failures. The restart is therefore the controlling lever at 2 qubits, a controlled contrast rather than a directional one. At 3 qubits, the per-stage restart across $n = 5$ seeds (`base_seed = 2001..2005`) reaches the correct basin on *all 5*, with a median of 4.0×10^{-3} and a tight range of 3.2–4.4×10^{-3}, insensitive to evaluation collocation density ($100 \approx 200$ points); the single-seed no-restart 1.7×10^{-3} reported in Table 2 was one such favorable draw, and no-restart 3q is correspondingly unreliable (1 of 3 in the initial screen). The consistent picture across qubit count is therefore: the plain combination is single-seed-optimistic and multi-seed-unreliable at 2–3 qubits, while the per-stage learning-rate restart delivers *reliable* closure and transfers up the ladder (2q 8/10 matched-seed, 3q 5/5, 4q 9/9 in-basin and 7/9 strict). There is no qubit wall—3q-with-restart remains the most reliable set (5/5, tightest spread), and the 4q median (8.8×10^{-4}, $n = 9$) is tighter still—but the honest headline is reliable closure via the restart, not the lucky-seed multipliers.

10 Code and Data Availability

Source code, run manifests, and the figure-generation pipeline are at `https://github.com/nhangen/DecoN`. Within the repository:

- Paper sources: `pre-print/arxiv-v3/`.
- Experiment manifests: `experiments/DeCoN-PINN/`.
- Figure-generation script: `scripts/paper/generate_arxiv_v3_figures.py`.
- Architectures (under `models/`): the {`One`,`Two`,`Three`,`Four`}`QubitUnitaryPINN` family, `OneQubitKrausPINN`, and `ChebyshevUnitaryPINN`.
- Trainer entry point: `scripts/train/safe_batch_trainer.py`; loss-modification options from Table 1 are exposed as CLI flags (`--curriculum-stages`, `--semigroup-weight`, `--terminal-pin-weight`, `--supervised-weight`, `--alm-initial-mu`, `--n-basis`).

11 Conclusion

The constraint-preserving PINN methodology of [2] generalizes to longer time intervals and closed-system multi-qubit dynamics with one consistent failure mode that we characterize as the local-vs-global gap: physics residual at N collocation points converges while the trajectory drifts. The gap is not closed by any of six loss-formulation modifications we tested. It is closed consistently across 1, 2, and 3 qubits by combining a structural extension of the architecture (unitary parameterization with structural IC anchoring) and a training-side modification (time-marching curriculum). Neither change alone suffices.

The original constraint-preserving result holds on the problem it solved; the gap is a scaling artifact at longer intervals and sustained-oscillation dynamics. Open-system curriculum methodology remains an open problem.

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